A RELATIVE PROJ OF A GRADED ALGEBRA GENERATED IN DEGREE ONE BY A FINITE TYPE MODULE IS PROJECTIVE OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-relative-proj-projective is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 7802-7810 and proof span 7812-7822 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof consists of the display of the graded algebra map and its citation, exactly as the author wrote it.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The construction of the morphism into a projective bundle associated with a degree one generating set of a graded algebra, which is a closed immersion when the symmetric algebra surjects onto the algebra, cited from the constructions chapter. Cited at proof lines 7812-7822. Produces the closed immersion of the relative Proj into the projective bundle of the degree one part, which is the claim.

Source reference. Original lemma 07ZK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10150–10150. Same-chapter reference used in the retained author text at source line(s) 7821, 8641. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{A}$ be a quasi-coherent graded $\mathcal{O}_S$-algebra. Let $p : X = \underline{Proj}_S(\mathcal{A}) \to S$ be the relative Proj of $\mathcal{A}$. If $\mathcal{A}$ is generated by $\mathcal{A}_1$ over $\mathcal{A}_0$ and $\mathcal{A}_1$ is a finite type $\mathcal{O}_S$-module, then p is projective.*

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Proof. Namely, the morphism associated to the graded $\mathcal{O}_S$-algebra map

$$\mathrm{Sym}^*_{\mathcal{O}_X}(\mathcal{A}_1) \longrightarrow \mathcal{A}$$

is a closed immersion $X \to \mathbf{P}(\mathcal{A}_1)$, see Constructions, Lemma 07ZK. □